

TECHNICAL ASSESSMENT

Text Summarization System: Design Brief

For the role of Natural Language Processing Engineer

ISSUED BY	SYNTHRA AI PVT. LTD., Language & Document Intelligence Group
ROLE	Natural Language Processing Engineer (Product Engineering)
DOMAIN	Legal & Compliance: Contract & Document Summarization
ASSESSMENT TYPE	Take-home design and implementation exercise
DATE OF ISSUE	30 June 2026
TIME ALLOWED	48 hours from receipt

1. CONTEXT

The candidate is being evaluated for a Natural Language Processing role at SYNTHRA AI, where engineering teams build production-grade summarization systems for legal, compliance, and contract-management platforms used by global product and platform clients. This assessment presents a problem representative of the work the team undertakes, and is intended to evaluate the candidate's ability to design, implement, and clearly communicate a working summarization solution.

2. PROBLEM STATEMENT

Design and implement a Text Summarization System that automatically generates concise, coherent, and faithful summaries of legal and compliance documents such as contracts, agreements, policies, and regulatory notices, condensing the source while preserving its key facts, parties, dates, obligations, and overall meaning.

3. OBJECTIVE

Given an input legal document written in the supported language and exhibiting one or more of the following characteristics:

- short-form or long-form text, ranging from a single clause or notice to a multi-page contract or master agreement;
- domain-specific or technical vocabulary, such as defined terms, parties, effective dates, monetary amounts, and legal terminology; or
- multiple sub-topics, cross-references, or shifts in focus within a single document,

the system shall produce a corresponding summary that is substantially shorter than the input, grammatically fluent, semantically faithful to the source, and suitable for direct use without further manual editing.

4. FUNCTIONAL REQUIREMENTS

4.1 Input and Length Support

The system shall correctly accept and summarise plain-text legal documents of variable length, from a single short clause or notice to a multi-page contract or agreement, and shall operate at a candidate-stated and justified compression ratio.

4.2 Core Architecture

The system shall be built around a Transformer-based encoder–decoder architecture. Candidates are required to demonstrate, and justify in their explanation, the role of each of the following mechanisms within their design:

- **Self-Attention**, for capturing dependencies between tokens within each sequence;
- **Multi-Head Attention**, for jointly attending to information from multiple representation subspaces;
- **Positional Encoding**, to preserve word-order information in the absence of recurrence or convolution;
- **Transfer Learning**, through the use of a pretrained language model (e.g., a BERT-family encoder for extractive summarisation, or a pretrained encoder–decoder checkpoint such as BART, T5, or PEGASUS) fine-tuned for the summarization task.

4.3 Processing Pipeline

The system shall accept an arbitrary input document, perform the necessary preprocessing, tokenisation, summarisation, and any necessary post-processing, and return a single summarised output without requiring additional manual intervention or document-type-specific templates.

4.4 Faithfulness and Intent Preservation

The system shall preserve the key facts, parties, dates, obligations, and intent of the source document. For example, monetary amounts, effective dates, defined terms, and named parties shall be carried through accurately and not distorted or fabricated. Because legal text is high-stakes, the summary must not introduce terms, obligations, or parties that are absent from the source, and shall reflect the emphasis and overall meaning of the document rather than a narrow or unrepresentative fragment of it.

4.5 Quality of Output

The output summary shall:

- be significantly reduced in length relative to the input, at a compression ratio stated and justified by the candidate;
- achieve a measurable standard of quality against an established automatic metric (e.g., ROUGE-1, ROUGE-2, ROUGE-L);
- read as fluent and natural, in a register appropriate to legal and compliance writing; and
- be free of hallucinated content not present in, or reasonably implied by, the source document.

4.6 Generalisation

The system shall not be tuned to, or restricted to, a single dataset or a narrow document type. It is expected to perform reasonably on previously unseen documents drawn from a variety of legal document types, jurisdictions, and drafting styles.

5. INPUT AND OUTPUT SPECIFICATION

Input	A single plain-text legal document in UTF-8 (e.g., a contract, agreement, policy, or compliance notice, or an extracted document body).
Output	The corresponding plain-text summary in UTF-8, of a length appropriate to the stated compression ratio.

6. EVALUATION CRITERIA

Submissions will be assessed against the following criteria. Candidates are advised to address each criterion explicitly in their submission.

6.1 Summarisation Quality	The coherence, fluency, and faithfulness of the generated summary, evaluated both qualitatively and using standard quantitative metrics (e.g., ROUGE-1, ROUGE-2, ROUGE-L) where appropriate. Particular weight is placed on factual faithfulness, given the legal context.
6.2 Robustness	The ability of the system to handle documents of varying length, structure, and legal drafting style.

6.3 Generalisation	Performance on document types, jurisdictions, and drafting styles not seen during development.
6.4 Architectural Soundness	The correct and well-justified use of attention mechanisms, encoder–decoder design, and transfer learning from a pretrained model.
6.5 Clarity of Approach	The quality, structure, and correctness of the candidate's explanation of their methodology, including the choice of extractive vs. abstractive summarisation.

7. SUBMISSION REQUIREMENTS

Candidates are required to submit, in addition to the working system, a short explanation video. The video is a mandatory component of this assessment and shall cover, at a minimum:

- the candidate's overall approach to the problem and the rationale for the chosen architecture, including the encoder–decoder design, the attention mechanism(s) used, and any pretrained model leveraged via transfer learning;
- a clear description of how the system works internally, including preprocessing, tokenisation, the summarisation model, and any post-processing; and
- a step-by-step dry run on at least one representative legal document, demonstrating how the input is transformed at each stage of the pipeline (e.g., tokenisation, attention/encoding, decoding, and summary generation).

Submissions that do not include the explanation video will be considered incomplete.

8. TIMELINE

The assessment shall be completed and submitted within forty-eight (48) hours from the time of issue. Late submissions will not be considered unless an extension has been agreed in writing in advance.

Note to candidate:

You are encouraged to use any publicly available libraries, frameworks, or pretrained models (e.g., Hugging Face Transformers, BART, T5, PEGASUS, BERT), provided that all such usage is disclosed in your explanation. Originality of approach is valued, but so is engineering judgement and clarity of communication. If any requirement in this document is unclear, please request clarification before commencing work.